Mini Project

Azure Cloud Fundamentals and Data Pipeline Implementation using Azure Data Factory

Celebal Technologies Internship

Submitted By

Assignment No-3: Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Dataset: Sample Superstore Dataset

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Technology: Microsoft Azure & Azure Data Factory

Problem Statement

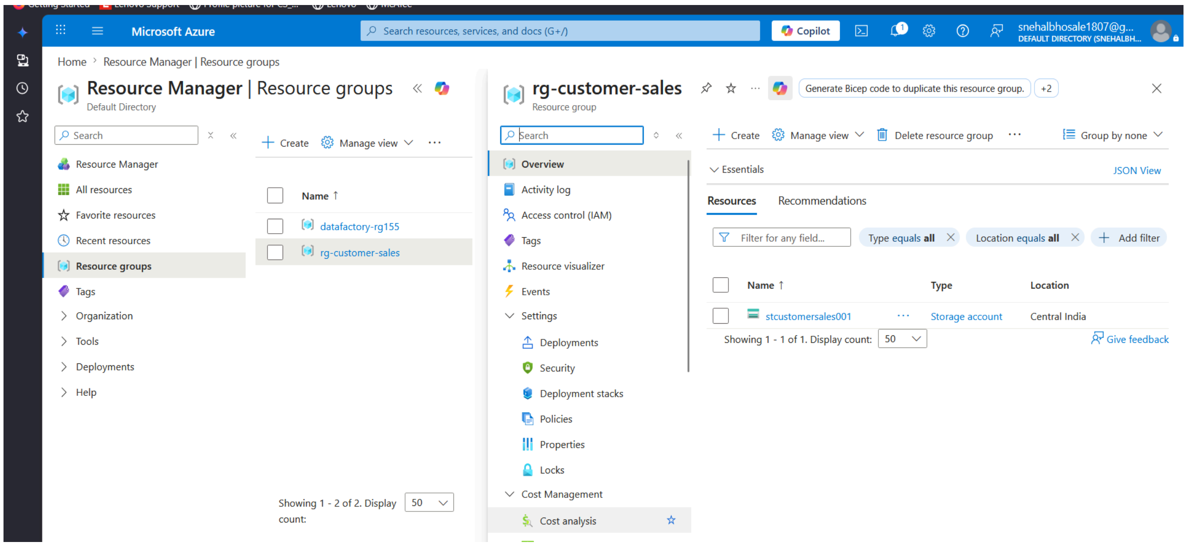
Build a complete data pipeline that reads a CSV file from Azure Blob Storage, validates the file using metadata, and copies it to a destination Blob container using Azure Data Factory.

Requirements

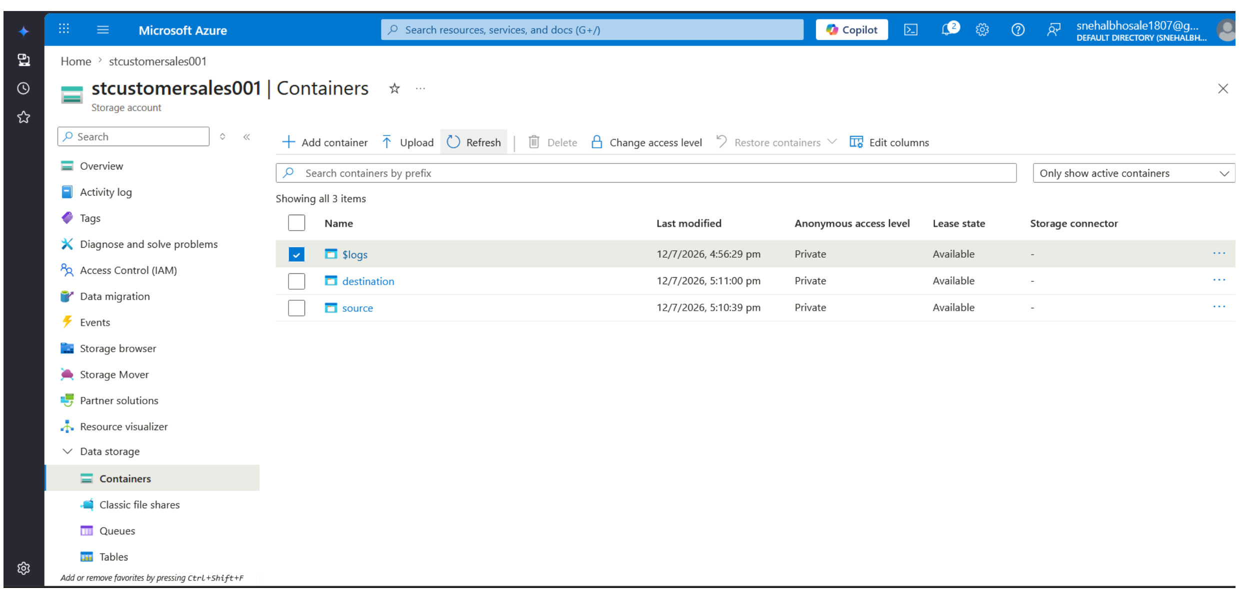
- **Source:** CSV file stored in the **source** Blob container.
- **Use:** Azure Blob Storage, Linked Service, Source Dataset, Destination Dataset, and Azure Data Factory Pipeline.
- **Process:** Validate the source file using **Get Metadata** activity and copy the data using **Copy Data** activity.
- **Destination:** Store the copied file as **Superstore_Output.csv** in the **destination** Blob container.

Implementation Steps

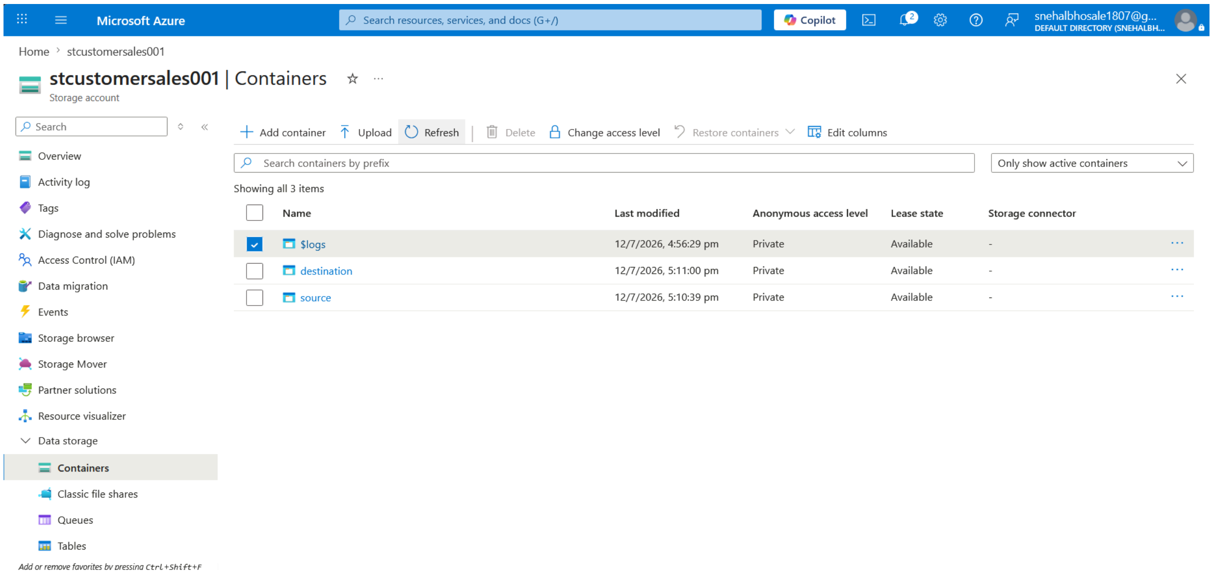
Step 1: Created a Resource Group (rg-customer-sales) to organize all Azure resources.



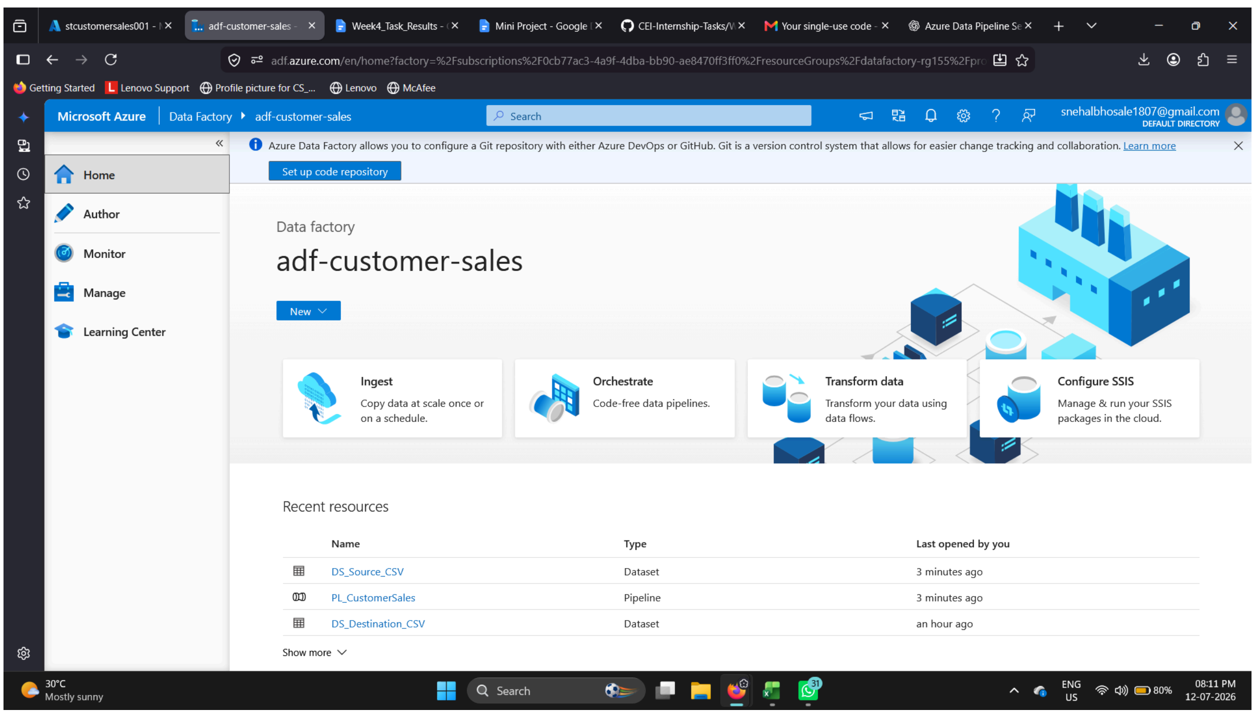
Step 2: Created a Storage Account (stcustomersales001) and two Blob containers named **source** and **destination**.



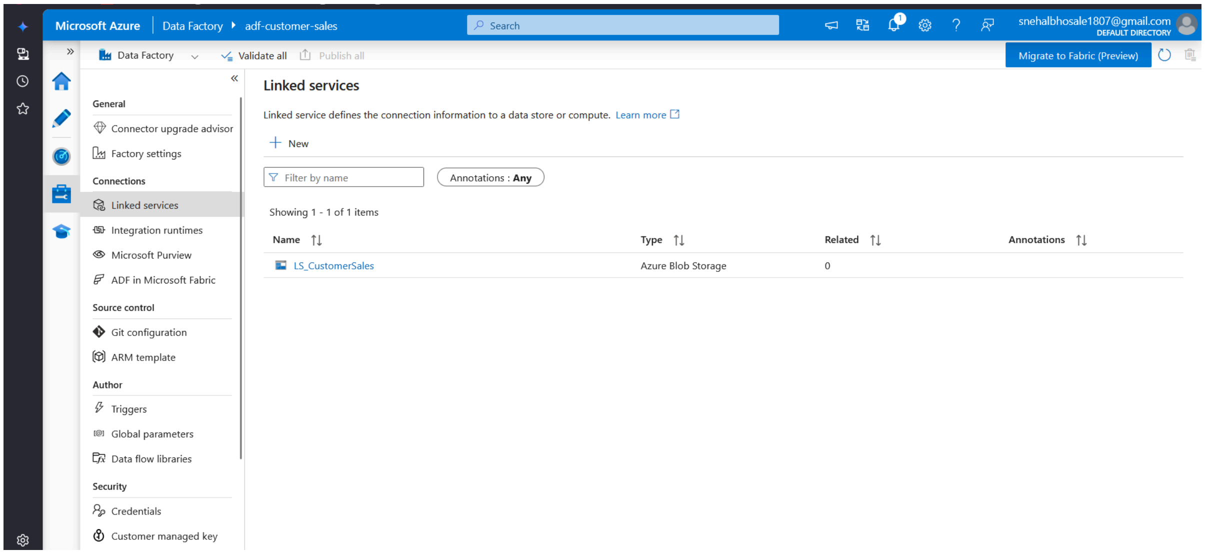
Step 3: Uploaded the **Sample - Superstore.csv** file into the **source** Blob container.



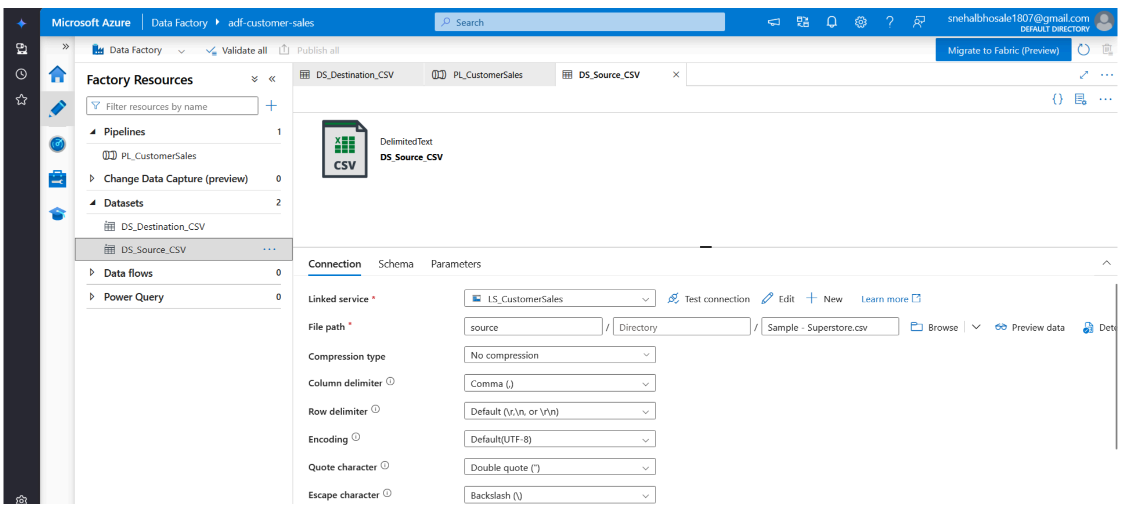
Step 4: Created an Azure Data Factory instance (adf-customer-sales) and launched ADF Studio.

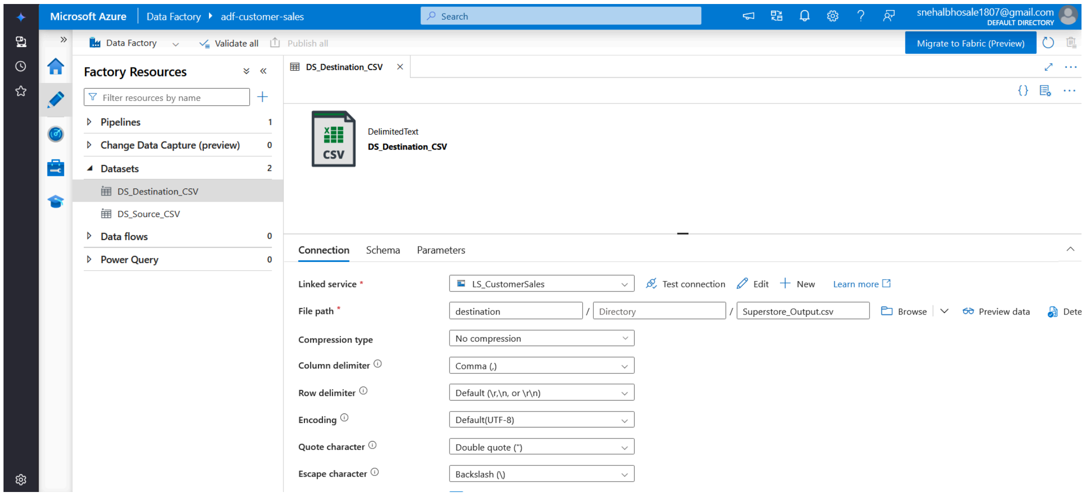


Step 5: Configured a Linked Service to establish a connection between Azure Data Factory and Azure Blob Storage.

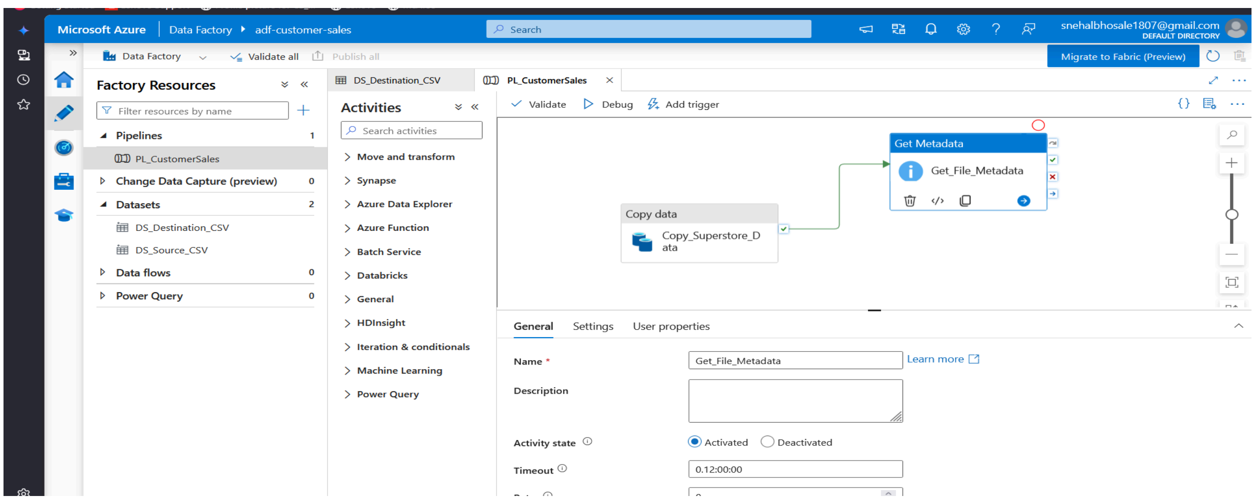


Step 6: Created the **Source Dataset** and **Destination Dataset** for reading and writing the CSV file.

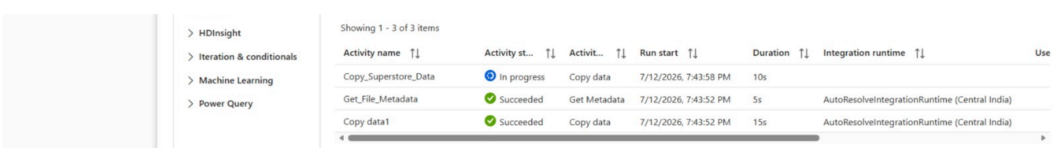




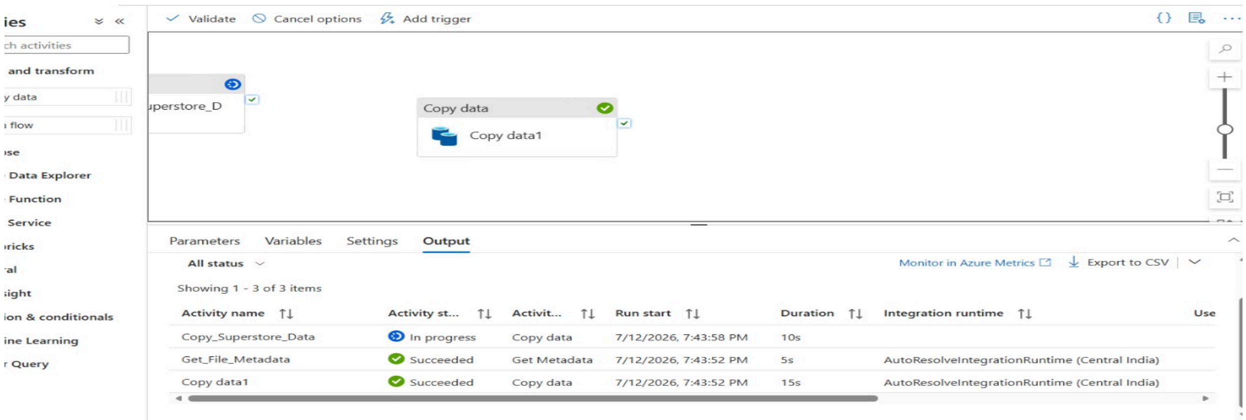
Step 7: Built a pipeline by adding **Get Metadata** and **Copy Data** activities.



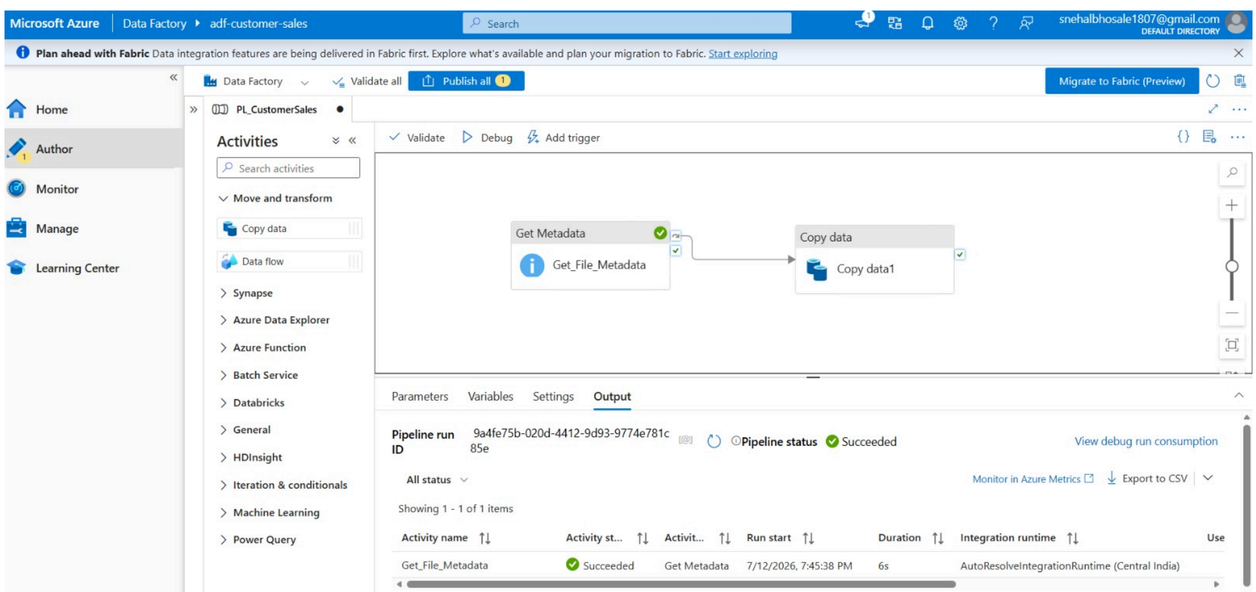
Step 8: Configured the **Get Metadata** activity to validate the source file by checking its existence, size, and last modified date.



Step 9: Configured the **Copy Data** activity to copy the CSV file from the **source** container to the **destination** container.



Step 10: Validated the pipeline, published all changes, executed the pipeline, monitored the execution, and verified the output file in the destination container.



Expected Output

- Pipeline executed successfully.
- CSV file copied from the **source** Blob container to the **destination** Blob container.
- Source file metadata validated successfully.
- Output file (Superstore_Output.csv) generated in the destination location.
- Pipeline execution monitored successfully in Azure Data Factory.

Learning Outcome

- Understood the basics of Microsoft Azure cloud services.
- Learned to create and manage Resource Groups and Storage Accounts.
- Gained hands-on experience with Azure Blob Storage.
- Learned to configure Linked Services and Datasets in Azure Data Factory.
- Built and executed an end-to-end data pipeline.
- Used **Get Metadata** and **Copy Data** activities for data processing.
- Learned to validate, publish, monitor, and troubleshoot pipelines.
- Understood the role of Azure IAM in providing secure access to cloud resources.

Conclusion

This mini project successfully demonstrated the implementation of an end-to-end data pipeline using Azure Blob Storage and Azure Data Factory. The pipeline validated the source file using metadata, copied the CSV file to the destination container, and completed the execution successfully. This project provided practical experience in cloud-based data integration and strengthened my understanding of Azure Data Factory and modern data engineering concepts.